Beyond One Disease: The Struggle to Prove the Value of Healthy Aging

In November of 2025, the 12-year-long cooperation between Calico Lab and pharmaceutical tycoon AbbVie, spending 1.75 billion dollars on the development of anti-aging drugs with zero outcome, came to an end. With such top resources, failure is still inevitable. Indeed, during the past century, while human beings' lifetimes were lengthened, new challenges of dealing with aging problems during prolonged old age were brought about. Recognition of the challenge has arisen among patients and health departments, while the direction of advanced biomedical innovation remains largely unchanged, still focusing on the examination and treatment of discrete diseases. In fact, only therapies that correspond to specific problems are granted opportunities for further development and clinical trials. On the contrary, being hard to define, evaluate, and commercialize by regular standards, multi-system declines are considered to be less important, even though they ultimately determine whether older people can maintain functional capacity. Currently, a newly burgeoning technique—synthetic biology—provides a regulatory pathway that largely meets the intervention standards for multi-system declines. Facing undefinable age-related health issues, it is able to deal with them through automatic and precise manipulation of intracorporeal therapy processes by building special circuits. However, without changing the innovation and investment direction from profitable diseases, synthetic biology's technical capacity may only become another contributing element towards the direction it means to prevent.

The age-related decline in the body system usually shows as multiple diseases. This is because, normally, to respond to cellular damage, the human body regulates these damaged cells to stop dividing, a process called senescence. As age grows, the human body gradually loses its ability to clear senescent cells efficiently. The accumulation of senescent cells will then damage surrounding tissues by releasing inflammatory signals. Any treatment targeting cellular senescence must therefore clear tissue-damaging senescent cells while preserving the short-lived senescent cells that support repair. In 2024, researchers built synthetic protein circuits that triggered a selected cell-death program only when they detected a specific combination of proteases. The utilization of cell signals, in this case, reduced the chance of targeting the wrong cells based on a single marker. More complex circuits can also respond to

changes over time. A feedback circuit responds to changes in the body. It controls the treatments' intensity according to the level of abnormal signals. The built circuits make use of and become part of the patients' own body system, so that the therapy schedules can be determined by patients' own biological state properly. Medicine could intervene when circuits detect the failure of patients' own regulatory system, protecting the tissues from irreversible damage. As the main therapeutic goal, declining body functions are preserved by repair and complement with circuits. Targeting early biological decline before it matures into a distinct disease, synthetic biology might safeguard function, as the affected tissue still retains some recoverability.

The WHO's ICOPE approach already asks health systems to organize care around intrinsic capacity and functional ability. Synthetic biology could give that model a technical basis at the level of cells to regulate biological failures that cross diseases and change over time. But in practice, international trial standards require a defined population and a measurable endpoint, which "better aging" is clearly unfit for. Under ICH E9(R1), the population, outcome, and handling of events such as death or treatment withdrawal are predefined by sponsors. The FDA also expects a clinical outcome assessment to be fit for a specified use. A healthspan trial would need to connect a change in capacity to something patients experience, like delayed disability or reduced dependence. This may take years to prove. Older participants' diseases are always variable and developing, as are their treatments, making it unreasonable and impossible to attribute certain effects solely to one circuit. A composite score, however, may also overlook or overestimate the actual effect of certain treatments, as it is obtained from a combination of therapeutic effects. A narrow disease indication makes the first trial easier to run, like senolytics, which were first tested in idiopathic pulmonary fibrosis. By doing so, researchers could enroll a defined group and measure safety and physical function within a shorter period, which also decides where evidence begins to form.

One trial called Targeting Aging with Metformin, or TAME, provides a way to test senescence treatment, incorporating multiple age-related diseases in one test. The trial's initial design is to recruit 3,000 older adults, from 65 to 79 years old, at 14 discrete sites, then observe their health condition during a six-year metformin

treatment, with the primary endpoint of cardiovascular disease, cancer or dementia progression, or death. By counting observed data with a strong relationship as one, researchers could obtain a set of statistically usable event rates with fewer participants and within a shorter time. This statistical approach was officially approved by FDA guidance for its advantages. However, designed to give an integrated result, TAME results can neither narrow an effect on a certain disease nor promise the preservation of body system capacity. Although metformin gives TAME a favorably safe case with a long medication history and low cost, AFAR is still struggling to raise funds for the trial of generic drugs with low profit. Similar or harsher requirements will be set against synthetic circuits. Given the long-lasting persistent nature of circuit treatment, FDA guidance may require extended monitoring, forcing regulators to weigh uncertain composite benefit against less reversible safety risk.

An initial indication of curative effect is also needed for the progression of a senescence circuit. Once a disease provides an indication, trials and capital it attracts would boost the investigation, much faster than testing a multi-system decline. Synthetic biology also takes the same path by narrowing the treatment while not losing its original direction in disease markets. Unless regulators accept preserved function as evidence of benefit and health systems reward therapies that sustain it, healthspan will remain outside the decisions that determine which circuits are built.

As a result, to collect standardized evidence, healthspan needs another path. The establishment of safety initially in an indication could enable the measurement of preservation of intrinsic capacity among patients with different diagnoses. This method can therefore generate usable results while not considering senescence as a typical disease. In the market where the observation period is prolonged and gains accrue partly to families and health systems, public funding may be necessary depending on payers' evaluations of their benefits. Moreover, the world beyond trial markets is also affected. As projected by the WHO, 80 percent of old people will live in low- or middle-income countries by 2050. As long as the major markets still set the standards and build central circuits around their own diseases and profits, countries that face the fastest aging and have lower therapy manufacturing and monitoring capacity will be the last to receive the therapies. Although synthetic biology enables therapies to have finer controls technically, the technique itself can't decide the

direction of investment. Those decisions are still made by the funding institutions. Only when the institutions measure a circuit by its maintenance of function and extend its value beyond the market that first proved it, could healthspan then become a global medical goal.